CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester Master of Technology (CE)
University Examination January 2011
CE703: Algorithm Complexity and Computations

Date: 05.01.2011, Wednesday **Time: 10:30 a.m to 1:30 p.m** **Maximum Marks:70**

Instructions:

1. *Attempt all questions.*
2. *Answers both sections in separate answer books.*
3. *Figure to the right indicates full marks.*
4. *Assume suitable data, if necessary and mention it.*

SECTION-I

Q-1

[A] What is an algorithm? What are the characteristics of an algorithm? 2

[B] On what kind of input does the Quick sort algorithm exhibit its worst-case behavior? Why? 2

[C] What are spanning tree and minimum spanning tree (MST) of a graph? 2

[D] Determine the minimum number of comparisons needed to compute the minimum and the maximum of n numbers 2

[E] How selection sort is better than bubble sort in the case when array is given into reverse sorted order? Explain with the help of a suitable example. 3

Q-2

[A] Solve the following recurrence exactly. 6

$t_n = 0$ if n=0

$= 1 / (4 - t_{n-1})$ otherwise

[B] What is the Greedy approach? Does it always give Optimal Solution? Give two examples in which Greedy algorithm gives optimal solution. 3

[C] Why in heap sort we need a left justified tree and not a right justified tree? 3

OR

Q-2

[A] Define 0/1 Knapsack Problem. Solve the following given instance Using Greedy Approach. Is this solution Optimal? Also solve using Backtracking approach and compare it with greedy approach in terms of time and space. 6

Capacity of Knapsack is 50.

Item	Weight	Value
1	10	60
2	20	100
3	30	120

[B] Is dynamic programming a Top-Down or a Bottom-Up technique? Why? Explain with an example. 3

[C] Explain how parallel bubble sort increases efficiency of bubble sort with suitable example. 3

Q-3

[A] Develop an algorithm using recursive and iterative function to calculate the GCD of two integers A and B. Also Find the Best Case and Worst Case time complexity of both functions. Comment on your answer. 4

[B] Explain the various asymptotic notations used in algorithm design. 5

[C] Is it possible to run radix sort on multiprocessor system? If yes then how it is possible. 3

OR

[C] Give an algorithm that determines whether or not a given undirected graph G=(V, E) contains a cycle. 3

SECTION-II

Q-4

[A] State Master's Theorem. 2

[B] What do you mean by backtracking and why is it required? Why is it so called? 2

[C] Differentiate among P, NP, NP-complete and NP-hard class of problems with suitable examples. 2

[D] What is a feasible solution and what is an optimal solution? Give suitable example. 2

[E] What are the issues that govern or influence the choice of either adjacency matrix or an adjacency list for storage of graphs? 3

Q-5

[A] Explain Chain matrix multiplication. Derive its time complexity. Why this is better than ordinary matrix multiplication? 5

[B] What is polynomial time reducibility? Give example. 2

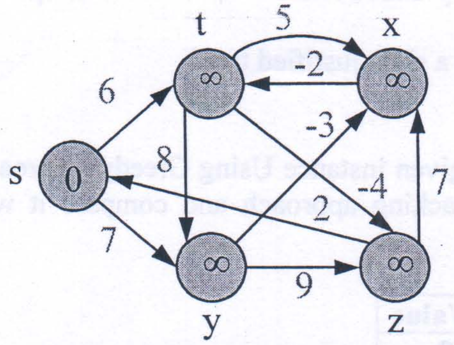
[C] Use branch and bound to solve the assignment problem (Minimization) with the following cost matrices. 5

Job	1	2	3	4
Person a	94	1	54	68
Person b	74	10	88	82
Person c	62	88	8	76
Person d	11	74	81	21

OR

Q-5

[A] Using Bellman Ford's algorithm, solve the single source shortest paths on a following weighted directed graph. The source vertex is indicated by 0. 5



[B] Compare Backtracking & Branch and Bound techniques with an example. 4

[C] Suppose we are comparing implementations of insertion sort and merge sort on the same machine. For inputs of size n, insertion sort runs in $8n^2$ steps, while merge sort runs in 64n lg n steps. For which value(s) of n does insertion sort beat merge sort? 3

Q-6

[A] Define Binary Search. What are the applications of binary search? State advantages & disadvantages of binary search. 4

[B] Write the Finite automata algorithm for string matching. Discuss its best and worst cases. 4

[C] Write an algorithm to compute the k^{th} smallest element of a list of n numbers, where k <=n. Determine the number of comparisons required to compute it, and deduce the time complexity of your algorithm. 4

OR

[C] Show that traveling salesman problem is NP complete. 4